

OC SurgeWatch: Bayesian Modeling of Surge Capacity for Orange County Hospitals – Health Disaster Preparedness

Joe T. Nguyen
UC Irvine, Master of Data Science
Bayesian Statistics STATS 205P
Spring 2026



Background & Motivation

- Orange County Health Care Agency (HCA) released its Hospital Surge Plan in June 2015, highlighting the **top hazards are wildfires, earthquake, and pandemic**
- During the initial response (0-72 hrs), an Impact Analysis is conducted:
 - Evaluate number of new and existing patients in the healthcare system
 - Assess the status of existing infrastructure
 - Determine level of unmet need for medical resources and patients
 - **Determine the capacity of known Emergency Receiving Center (ERC) hospitals, and their ability to receive the expected number of patients**
 - Determine the situational status of the health and medical system
- ***What if there was a way to predict which of these ERC hospitals are most vulnerable to a surge during a disaster?***



Overall Objectives

- Effective as of 1/1/2026, there are 25 Hospital Emergency Receiving Centers in OC
- Of these 25 facilities, I want to know:
 - *Which are most vulnerable to patient volume surges?*
 - *What facility-level characteristics could predict resilience versus fragility under a surge scenario?*



Orange County EMS Agency Policy/Procedure

LIST OF EMERGENCY RECEIVING CENTERS



#230.00
Page 1 of 1

HOSPITAL EMERGENCY RECEIVING CENTERS

1	Anaheim Global Medical Center
2	Anaheim Regional Medical Center
3	Rady Children's Hospital Orange County (Pediatric Patients)
4	Foothill Regional Medical Center
5	Garden Grove Hospital and Medical Center
6	Hoag Hospital – Irvine
7	Hoag Hospital – Newport Beach
8	Huntington Beach Hospital
9	Kaiser Permanente Medical Center – Anaheim
10	Kaiser Permanente Medical Center - Irvine
11	La Palma Intercommunity Hospital
12	Mission Hospital – Mission Viejo
13	Mission Hospital – Laguna Beach
14	Orange Coast Memorial Medical Center
15	Orange County Global Medical Center
16	Saddleback Memorial Medical Center
17	Saint Joseph Hospital
18	Saint Jude Medical Center
19	South Coast Global Medical Center
20	UCI Health – Fountain Valley
21	UCI Health – Irvine
22	UCI Health – Los Alamitos
23	UCI Health – Orange
24	UCI Health – Placentia Linda
25	West Anaheim Medical Center

Approved:



Carl H. Schultz, MD
OCEMS Medical Director



Michael Noone, NRP
OCEMS Assistant Administrator

Original Date: 1/1984

Reviewed Date(s): 6/12/2015; 6/1/2016; 05/22/2019

Revised Date(s): 4/1/2013; 6/12/2015; 6/1/2016; 05/22/2019; 09/23/2024; 12/16/2025

Effective Date: 1/1/2026

OCEMS Policy #230.00

Effective Date: 1/1/2026

Data Description

- **Data Source:** California Department of Health Care Assess & Information (HCAI) / Hospital Inpatient
 - Years: 2018 - 2024 (7 years)
 - Scope: Orange County ERC hospitals – 21 facilities ⇒ **147 data points (annual aggregates)**
 - 2 report under parent, 1 pediatric-only (structurally incomparable), 1 opened in 2025

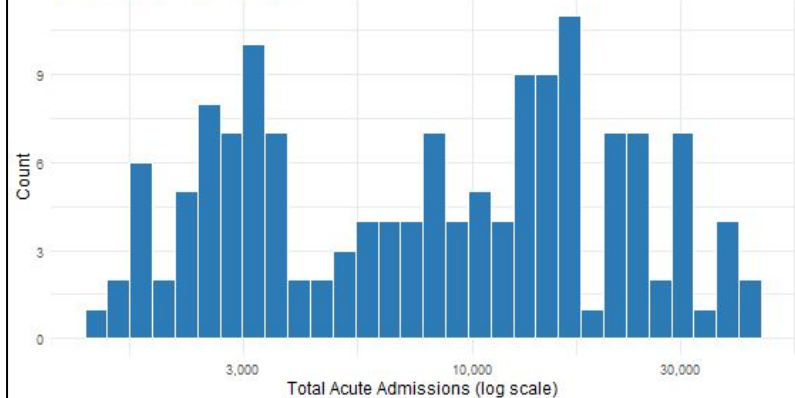
Variable Group	Variables Used	Modeling Role
Facility ID	oshpd_id, facility_name	Unit Identifier
Year	derived: from file year	Time Proxy
Facility Type	licensed_bed_size	Capacity
Discharge Days	dsch_days	Average Length-of-stay Proxy
Total Annual Admissions	toc_acute	Primary Outcome
Medicare Volume	payer_medicare	Scorecard: Financial Fragility
60+ Cohort	age_60_69, age_70_79, age_80_	Scorecard: Vulnerable Population
Emergency Admissions	adm_emergency	Scorecard: Surge Sensitivity
Occupancy Rate	derived: (dsch_days) / (licensed_bed_size x 365)	Scorecard: Capacity Strain

Let's Explore The Data

1

Distribution of Annual Acute Admissions (toc_acute - log scale)

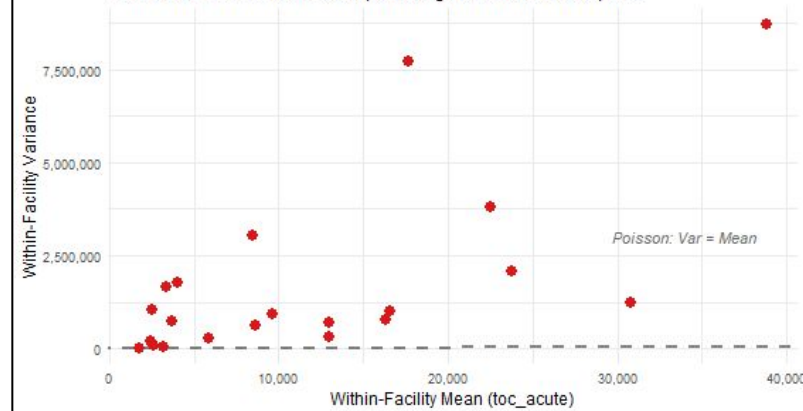
Bimodality on log scale — two facility sizes, not a single population.
Random intercepts may be needed.



2

Variance vs. Mean by Facility Across Years (Overdispersion Check)

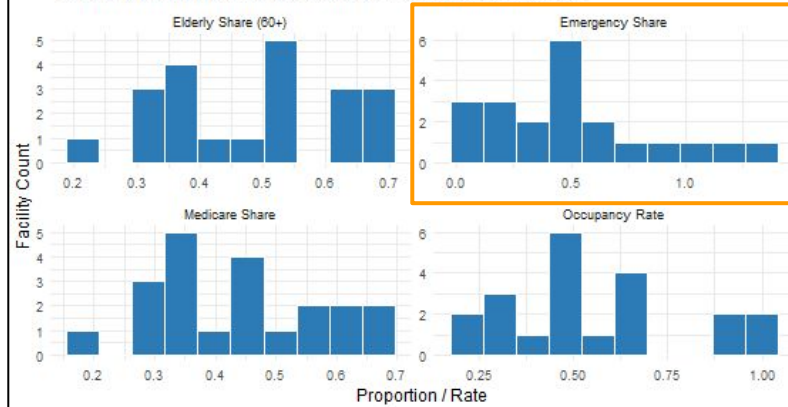
Poisson line is invisible at this scale — annual swings push observed variance way above the Poisson mean assumption. Negative Binomial is required.



3

Scorecard Feature Distributions (Baseline, 2018–2019)

21 facilities. Each scorecard share is a proportion of total admissions;
Reporting anomaly in Emergency Share exceeding total admissions.



4

Scorecard Feature Correlation Matrix

Elderly Share and Medicare Share are nearly collinear ($r = 0.92$).
Occupancy is moderately negative with all three ($r \approx -0.57$).



Modeling Implications

1

A hierarchical/mixture modeling approach that captures within-facility random intercepts — **allowing for comparison of each facility against their own baseline** rather than group average.

2

Overdispersion in count data requires a specialized **Negative Binomial** model that absorbs the extra variance while avoiding overconfidence in predictions.

3

Emergency admissions cannot exceed total admissions. Some hospitals failed to report correctly so this scorecard cannot be used reliably. Imputation without domain expertise is unwise.

4

Medicare and 60+ population are nearly collinear which is expected since most elderly patients are disproportionately paying with Medicare. To avoid multicollinearity issues in our predict, **Medicare scorecard is removed** from modeling.

Breaking Down The Problem

Q1: Which are most vulnerable to patient volume surges?

If disaster strikes tomorrow, any hospital over 95% bed capacity is extremely vulnerable.

95% bed capacity = how many beds are occupied at a given moment



Data Limitations:

- Annual Aggregates
- Lack of granularity

$$ALOS_j = \frac{\text{Discharge Days}_j}{\text{Annual Admissions}_j}$$

$$\text{Daily Census}_j = \frac{\text{Annual Admissions}_j \times ALOS_j}{365}$$

$$\text{Occupancy Rate}_j = \frac{\text{Daily Census}_j}{\text{Licensed Beds}_j} = \frac{\text{Discharge Days}_j}{\text{Licensed Beds}_j \times 365}$$

$$\text{Capacity Threshold}_j = \frac{\text{Licensed Beds}_j \times 365 \times 0.95}{ALOS_j} = \frac{\text{Licensed Beds}_j \times 365 \times 0.95 \times \text{Annual Admissions}_j}{\text{Discharge Days}_j}$$

$$P(\text{Admissions}_j > \text{Capacity Threshold}_j) = P(\text{Occupancy Rate}_j > 0.95)$$



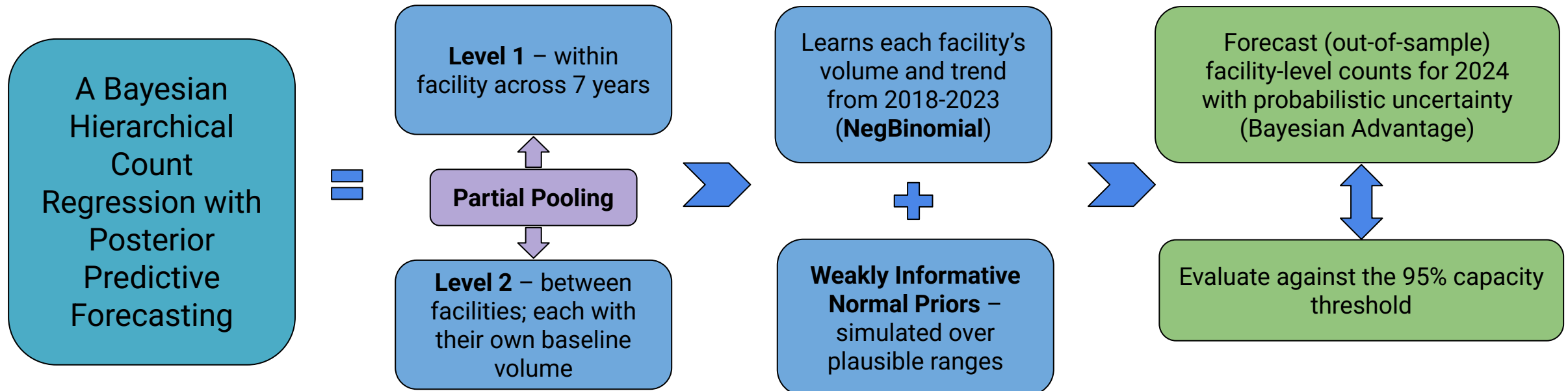
A Bayesian Hierarchical Count Regression with Posterior Predictive Forecasting

Ideally, real-time or weekly/monthly batch data would enhance our inference and predictive capabilities.

Modeling Architecture – M1

Q1: Which are most vulnerable to patient volume surges?

If disaster strikes tomorrow, any hospital over 95% bed capacity is extremely vulnerable.

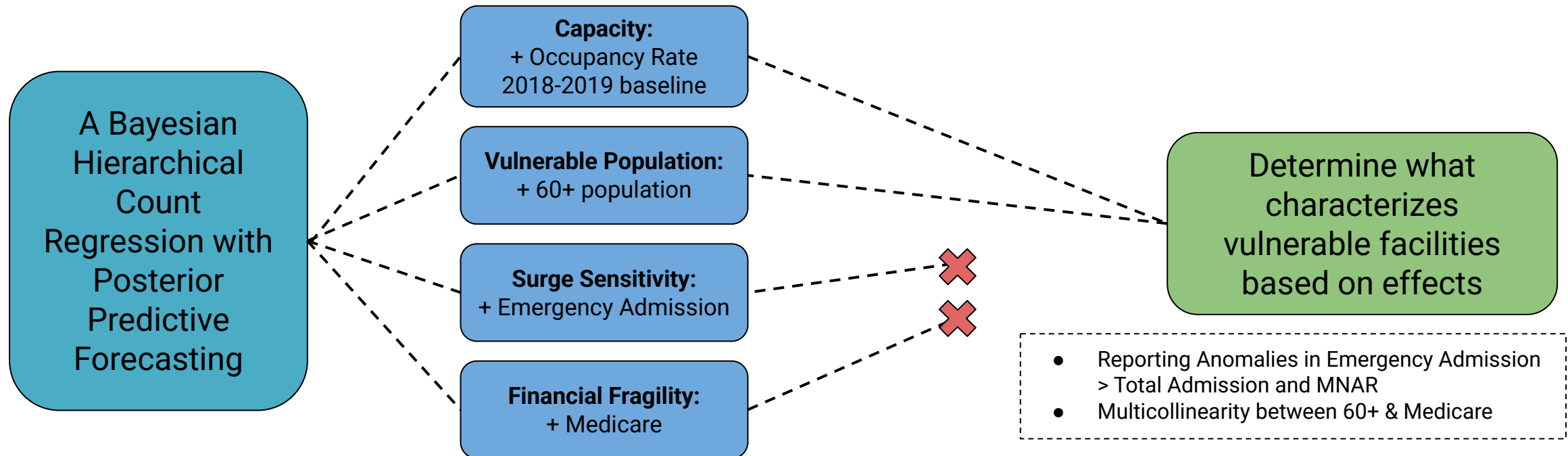


The resulting posterior predictive distribution ranks facilities by forecasted volume and probability of over 95% capacity.

Modeling Architecture – M2

Q2: What facility-level characteristics could predict resilience versus fragility under a surge scenario?

For each facility, can a model predict the future total admission count? If so, what features contribute most to the model's prediction?



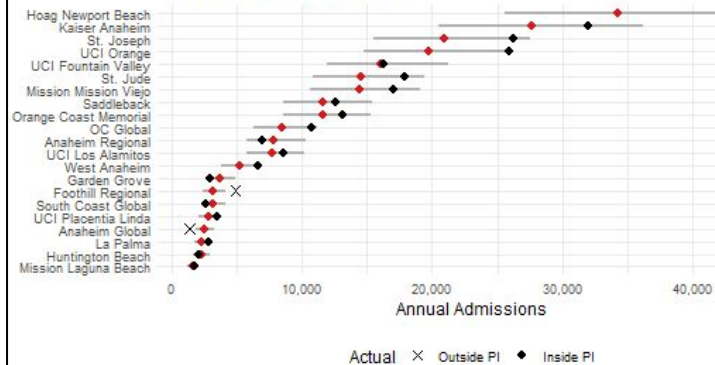
In a real-world scenario with full granularity, staffed beds, nurse ratios, ICU capacity, diversion rates, etc., would be more insightful characteristics.

Key Findings

1

Posterior Credible Intervals vs Actuals — Test Set (2024)

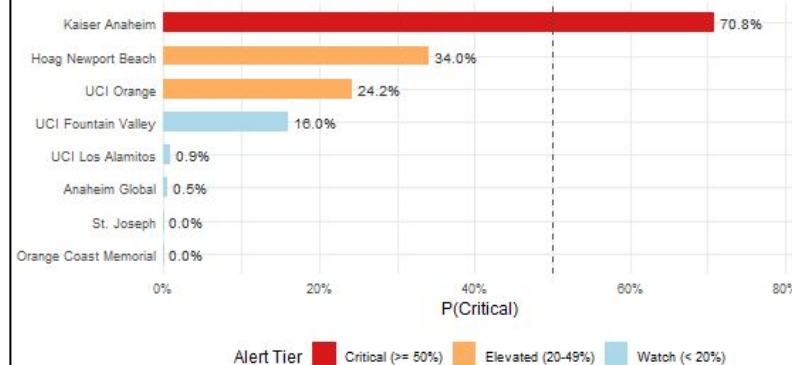
Red dot = predicted median; black dot = actual 2024 admissions.
Filled circle = actual falls inside 90% CI; X = outside.
Scorecard model, all 21 facilities.



2

Surge Risk Monitor — P(Exceeding 95% Capacity) by Facility

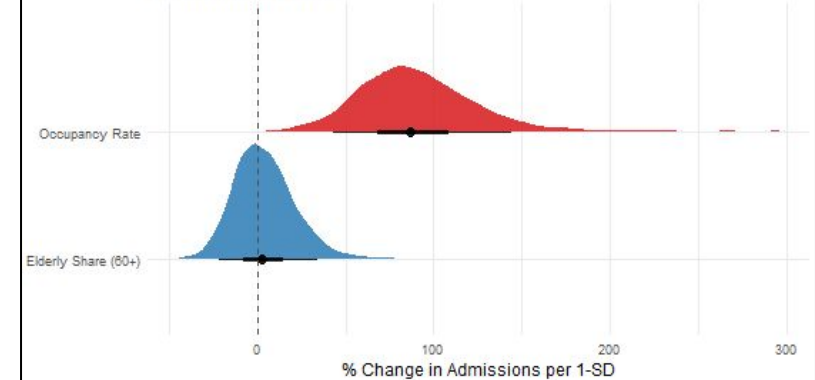
Probability that forecasted admissions exceed the facility-specific 95% bed occupancy threshold. Dashed line = 50% critical threshold. Only facilities with P(Critical) > 0 shown.



3

Scorecard Feature Effects on Annual Admissions

Posterior distribution of % change in admissions per 1-SD increase in each feature. Inner bar = 50% credible interval; outer bar = 90% credible interval. Dashed line = no effect.



1

The model is reliable for predicting mid-sized facilities but **underestimates demand at the largest facilities** — which are ironically the most critical facilities to get right in a disaster scenario. This is likely a consequence of the bimodal distribution. **85.7% of the 2024 actuals fell inside the baseline model's 90% posterior credible interval, where the scorecard model captured 90.5%.**

2

The facilities most likely to breach capacity are the ones already operating at high volume. In a disaster event where patient load redistributes across the system, **a facility already near its threshold has almost no buffer to absorb additional demand.**

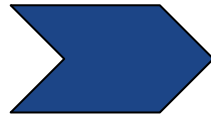
3

Baseline 2018-2019 occupancy rate carries meaningful predictive weight. Elderly (60+) share adds noise but not information. In a real-world monitoring system, pre-surge occupancy would be a strong facility-level metric to track.

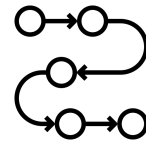
These findings are based on 2018-2024 annual aggregated data with a lack of granularity and should be interpreted carefully.

Proof of Concept with Real World Implications

Predict the capacity of known Emergency Receiving Center (ERC) hospitals and their ability to receive the expected number of patients during a hospital surge or disaster event.



Reinforce vulnerable ERC hospitals in preparation for disaster



Streamline capacity assessment during the initial 72-hour response window



Enable immediate decision-making and support coordination in crisis



Save lives!

Thank you! Questions?

